

JAVA
ADVANCED
INTERVIEW Q&A
PART - 1



@java_in.depth

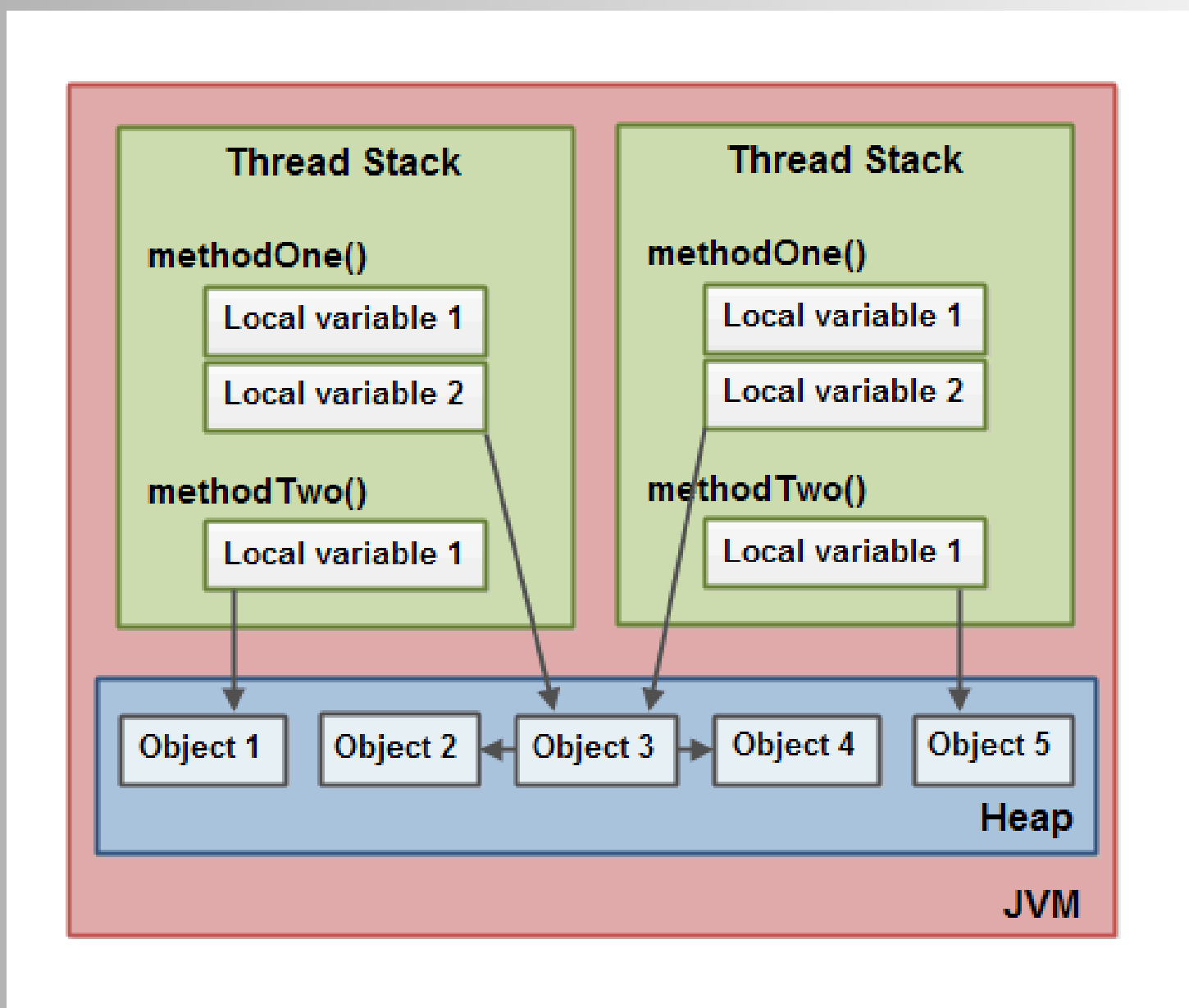
IMPORTANT JAVA INTERVIEW Q&A

Questions that frequently appear in senior
developer interviews

SAVE FOR YOUR NEXT INTERVIEW 🚀

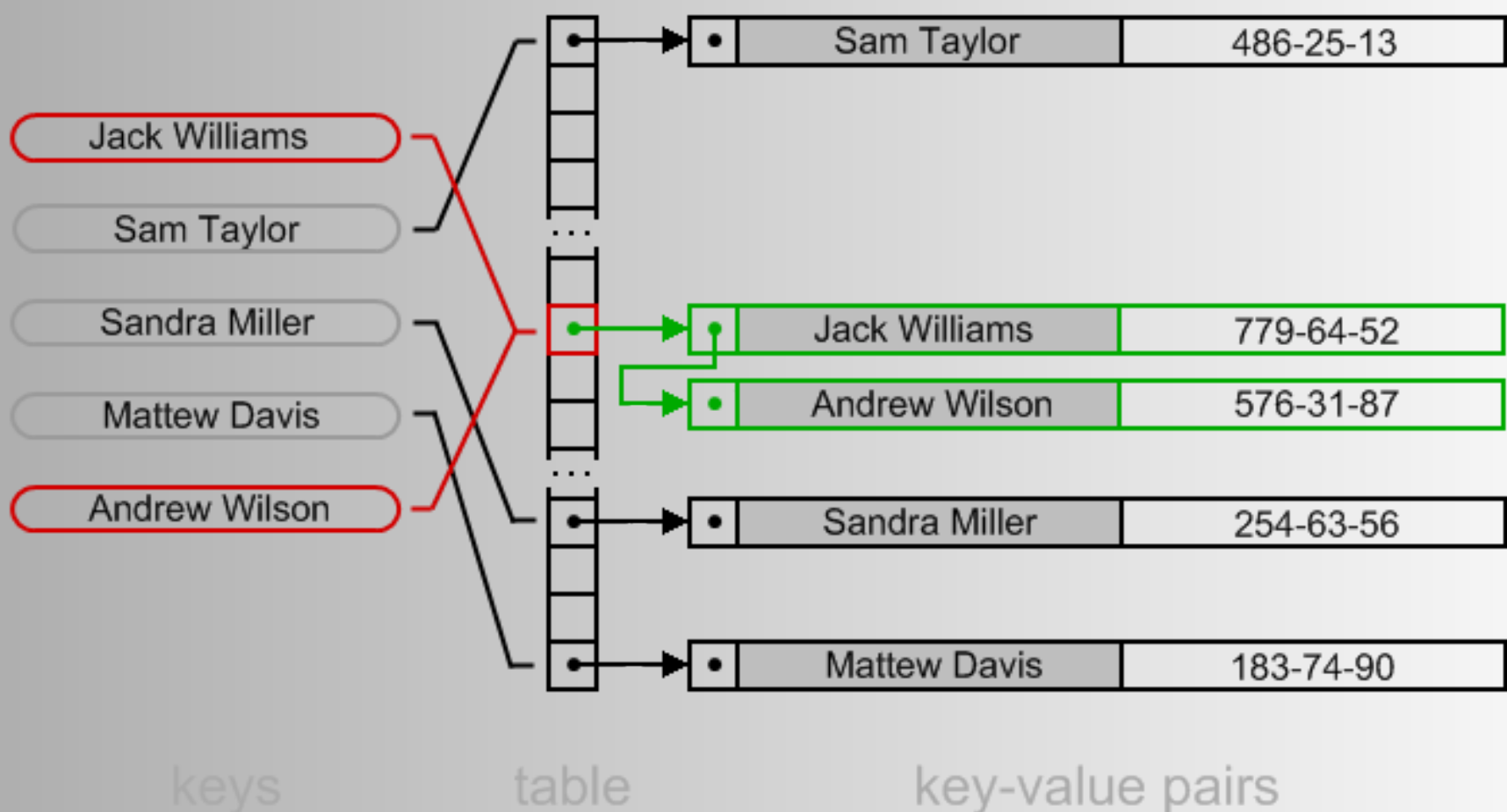
1. WHAT IS THE JAVA MEMORY MODEL?

The Java Memory Model defines how threads read and write shared data in Java. It ensures predictable behavior in concurrent programs by defining rules for visibility and ordering of operations.



2. WHY IS HASHMAP NOT THREAD-SAFE?

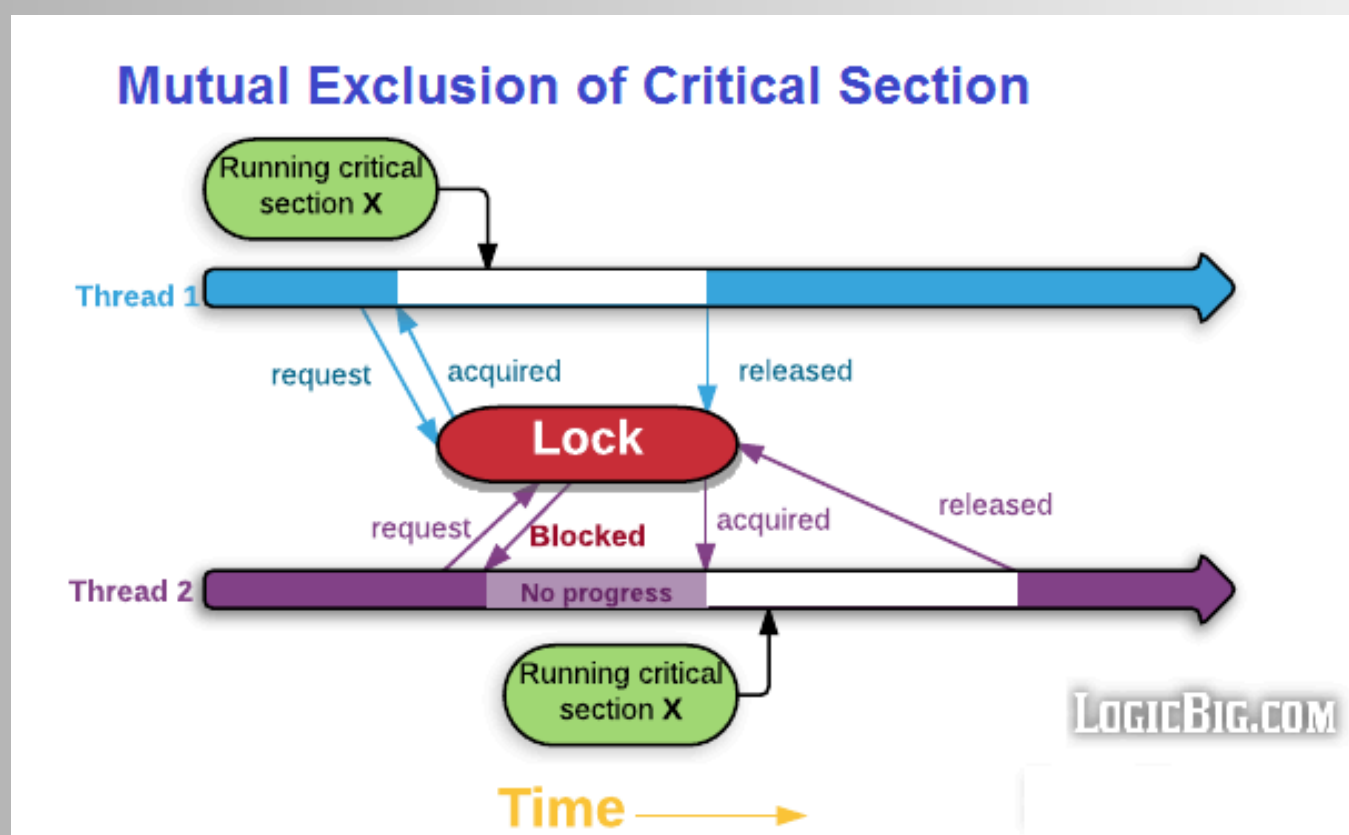
A HashMap allows multiple threads to access and modify data without synchronization. If two threads update the map simultaneously, it can lead to data corruption or inconsistent state.



3. DIFFERENCE BETWEEN SYNCHRONIZED AND LOCK

synchronized is a built-in mechanism that automatically acquires and releases a monitor lock. It is simple but less flexible.

Lock from the `java.util.concurrent` package provides advanced features such as `tryLock`, fairness policies, and interruptible locking, giving developers more control over concurrency.



4. WHAT IS THE DIFFERENCE BETWEEN STACK AND HEAP MEMORY?

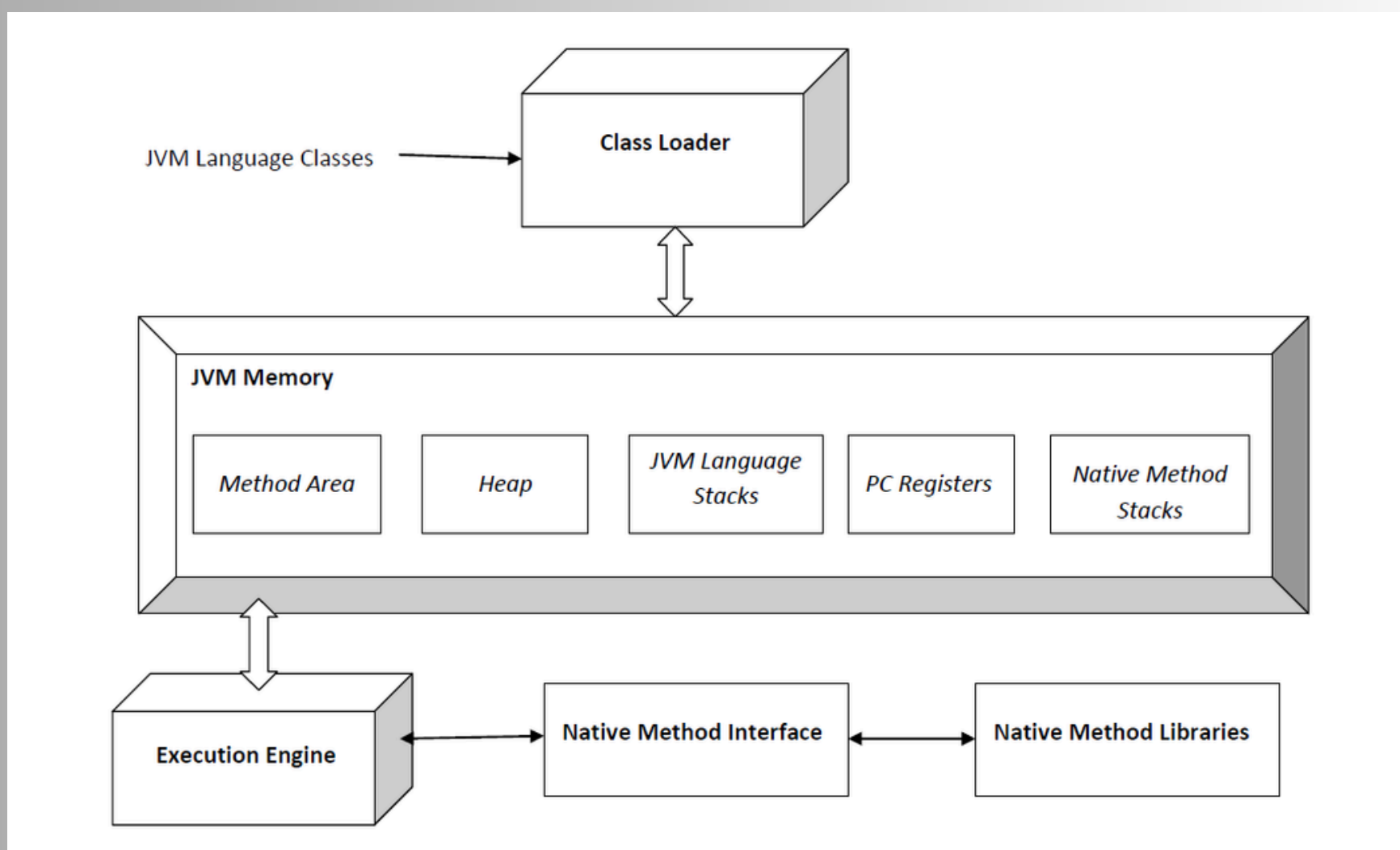
Stack memory stores method calls and local variables, and it is managed automatically when methods are executed.

Heap memory stores objects created during runtime. It is shared among threads and managed by the garbage collector.

Understanding this difference helps explain memory issues and performance behavior in Java applications.

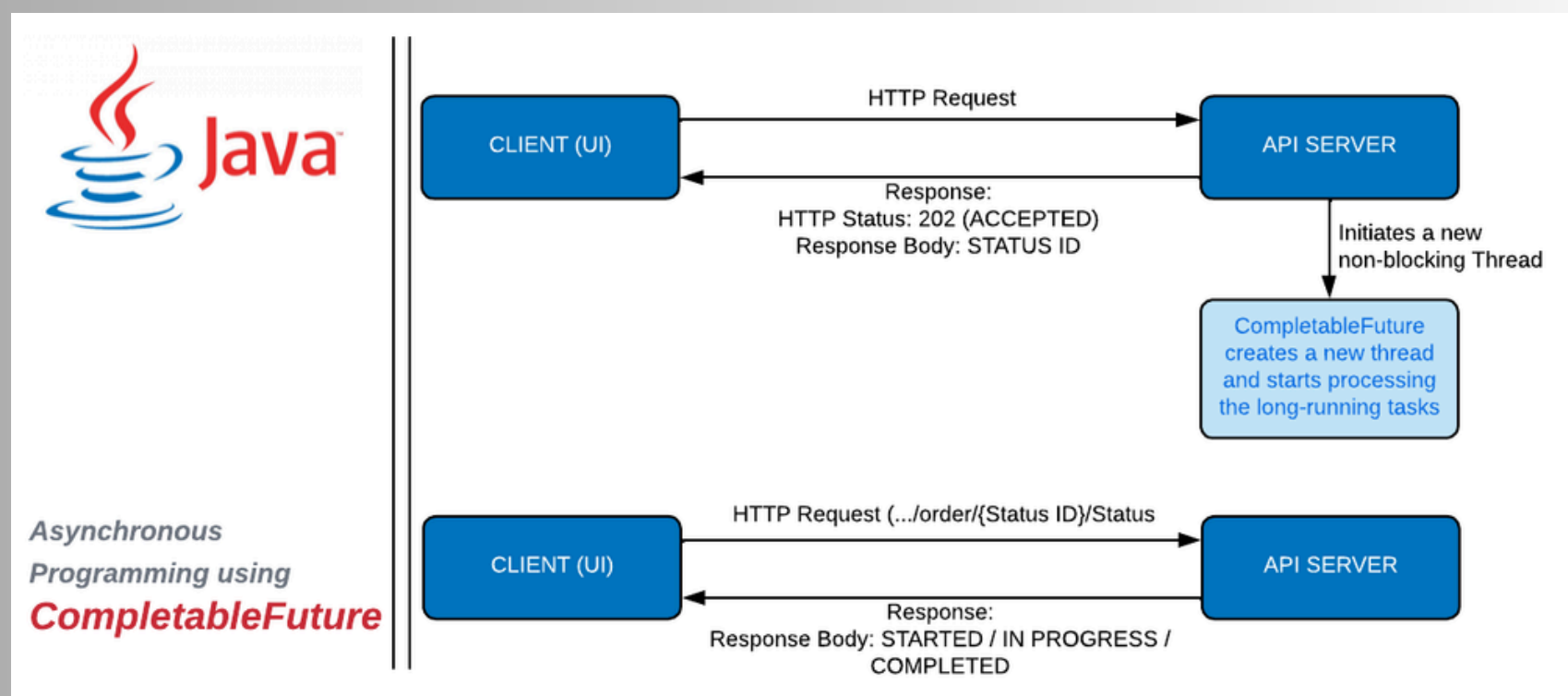
5. WHAT IS THE PURPOSE OF THE JVM?

The Java Virtual Machine executes compiled Java bytecode and provides platform independence. It includes components like the ClassLoader, execution engine, and runtime memory areas.



6. WHAT IS COMPLETABLEFUTURE IN JAVA?

CompletableFuture is used for asynchronous programming. It allows multiple tasks to run independently and then combine their results once completed.



7. WHAT IS REFLECTION IN JAVA?

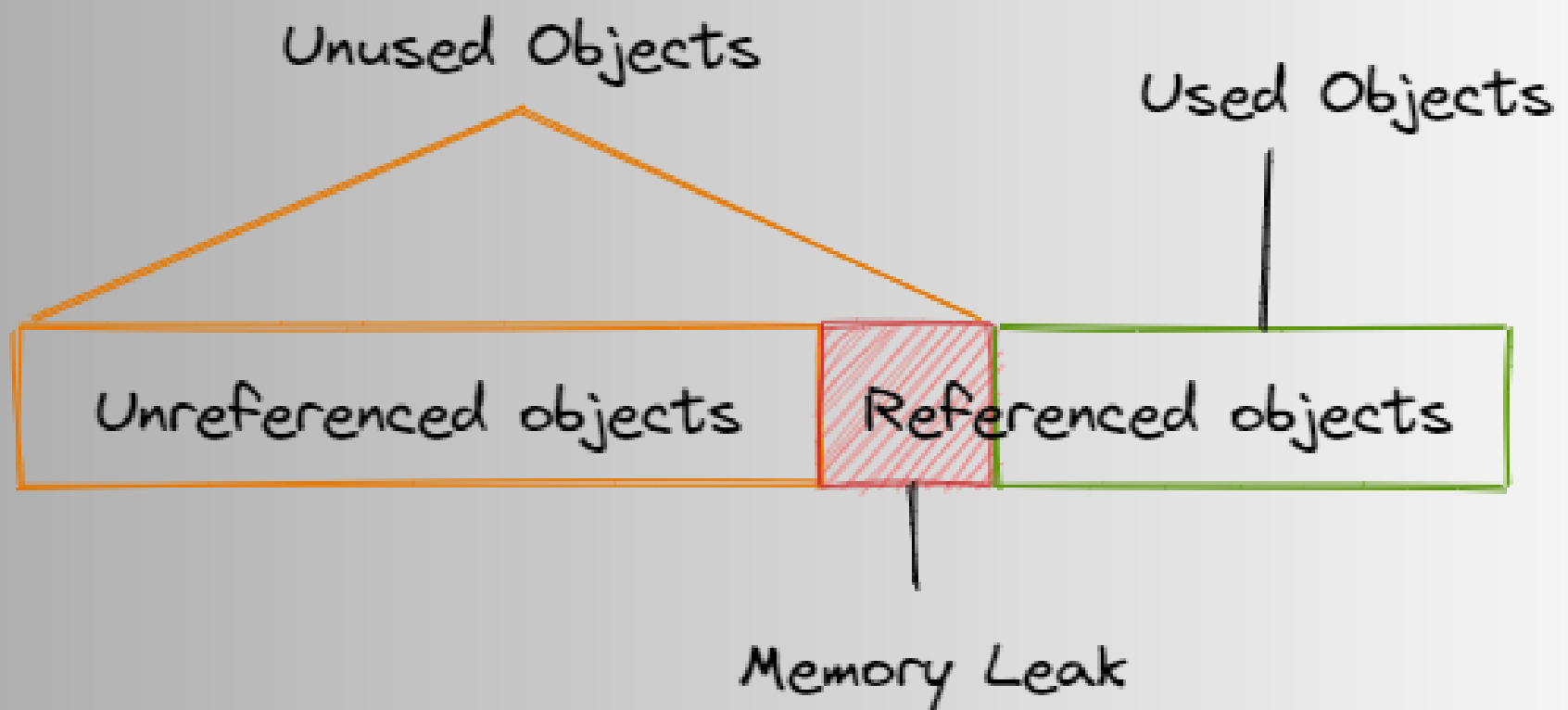
Reflection allows a program to inspect and modify classes, methods, and fields during runtime.

It is heavily used in frameworks such as Spring Framework where annotations and dependency injection rely on runtime inspection of classes.

8. WHAT CAUSES MEMORY LEAKS IN JAVA?

Even with garbage collection, memory leaks can occur when objects are still referenced but no longer needed.

Common causes include static collections holding objects, unclosed resources, and listeners that are never removed.



9. INTERVIEW TIP

In advanced Java interviews, companies focus more on understanding than memorization.

If you can clearly explain how memory works, how threads interact, and how the JVM executes programs, you already stand out from most candidates.



WANT PART 2? 📌

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